

STA-A

STA-B

Shared information: cipher suite, STA-A-MAC, STA-B-MAC
 If generating PWE by looping: *password*
 If generating PWE by hash-to-element: *PT*

Provide either *password* or *PT* to
`psa_pake_setup()` depending
 on PWE generation method

`psa_pake_setup()`
`psa_pake_set_user( STA-A-MAC )`
`psa_pake_set_peer( STA-B-MAC )`

Generate *rand, mask*; compute
commit-scalar, COMMIT-ELEMENT

`psa_pake_output()` for *commit-scalar* || *COMMIT-ELEMENT*

SAE Commit frame (*commit-scalar, COMMIT-ELEMENT*)

SAE Commit frame (*peer-commit-scalar, PEER-COMMIT-ELEMENT*)

Validate inputs; compute *k*

`psa_pake_input()` for *peer-commit-scalar* || *PEER-COMMIT-ELEMENT*

Compute SAE-KCK, PMK

`psa_pake_input()` for salt

loop

[Until SAE Confirm frame is successfully delivered to STA-B]

`psa_pake_input()` for *send-confirm* counter

Compute *confirm*

`psa_pake_output()` for *send-confirm* || *confirm*

SAE Confirm frame (*send-confirm, confirm*)

SAE Confirm frame (*peer-send-confirm, peer-confirm*)

Compute and validate
peer-verify = peer-confirm

`psa_pake_input()` for *peer-send-confirm* || *peer-confirm*

opt

`psa_pake_output()` for PMKID

`psa_pake_get_shared_key()` to extract PMK